Gemini 2.5 and its Computer Use model

Google's AI agent to navigate interfaces

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Google's latest Gemini 2.5 update has quietly introduced something that could reshape how artificial intelligence interacts with the web: the Computer Use model. Unlike traditional chatbots that merely generate text or code, this new capability allows Gemini to literally use a computer, clicking buttons, filling forms, scrolling through websites, and performing actions on-screen much like a human user would.

A new kind of "AI user"

The Gemini 2.5 Computer Use model is designed to bridge a long-standing gap between language models and the real world: the ability to interact directly with user interfaces.

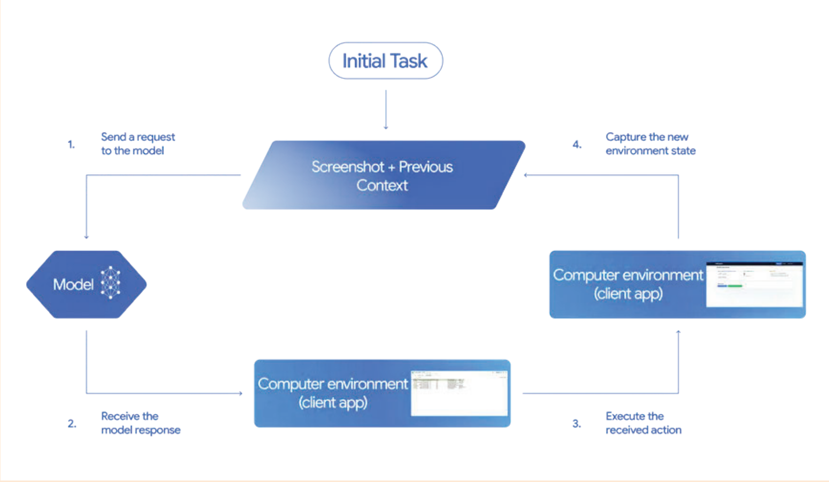
Traditionally, AI systems have relied on APIs to fetch information or perform tasks. But APIs aren't available for everything – especially the messy, visual world of websites and apps. Gemini's new model solves that by giving the AI eyes and hands in the digital sense. It can "see" what's on a screen through screenshots and then decide how to act: clicking buttons, typing into fields, or scrolling to find relevant sections.

The agent loop

At the heart of Gemini's Computer Use model is an iterative agent loop, a continuous process of perception, reasoning, and action.

1. **Observe:** The model receives a screenshot and contextual information about the current screen.
2. **Decide:** It analyzes the interface, determines the next logical step (like clicking a login button), and produces a function call describing that action.
3. **Act:** A client-side automation layer executes that action on the real interface.
4. **Repeat:** A new screenshot is captured, and the loop continues until the goal is met.

This system mimics how humans interact with computers, looking, thinking, and acting, only at machine speed and with higher consistency. Developers can even set safety constraints, preventing the AI from performing sensitive actions like confirming payments or sharing data without explicit approval.



What it can do

Google's early demonstrations showcase Gemini 2.5 handling real-world tasks. It can fill out online forms, navigate dashboards, rearrange elements in web apps, and even work across multiple tabs or sites to complete multi-step workflows. For instance, it could pull contact details from a website, paste them into a CRM tool, and then schedule a meeting in Google Calendar, all completely autonomously. This capability opens up a new frontier for business automation, testing web apps, data entry, and personal productivity.

Unlike traditional automation tools that depend on rigid scripts or APIs, Gemini's approach is adaptive. It "understands" visual layouts, making it more resilient to changes in a site's design or structure.

Safety first

A model that can control a computer naturally raises safety concerns. Google acknowledges this and has implemented multiple guardrails. Every proposed action is checked by a safety service before execution, filtering out anything harmful, malicious, or risky.

Certain actions, such as making financial transactions or sending data, require explicit user confirmation. Developers can also define "forbidden actions" to ensure the model stays within safe boundaries.

Moreover, Google recommends that all experiments with the Computer Use model take place in sandboxed environments, minimizing any risk of unintended side effects.

Currently, Gemini's Computer Use model is available only in preview through the Gemini API, accessible via Google AI Studio and Vertex AI.